

Photoionization Cross Sections in EXHALE:

H I, He I, He II, and the He I 2^3S Triplet,

with Benchmarks against Verner+1996 and Norcross 1971

Contents

1. Overview	1
2. The four cross sections	2
2.1 Hydrogenic atoms: H I and He II	2
2.2 He I ground state (1^1S)	2
2.3 He I 2^3S metastable (two VFXY96 forms + bridge)	3
2.4 Verner+1996 single-shell fit (VFXY96)	3
3. Overview comparison	3
4. He I ground state: the legacy two-term fit vs Verner+1996	4
5. He I 2^3S: the implemented cross section vs p-winds/Norcross	5
5.1 Cross-check against TOPbase / Opacity Project	5
5.2 Did the $> 59\text{eV}$ truncation matter for the rate?	7
5.3 Impact of the autoionizing resonances	8
5.4 A smooth high-energy extension ($E > 59\text{eV}$)	9
5.5 A unified full-range representation from TOPbase	10
5.6 A minimal-change extension of the broken power law	11
5.7 He 2^3S as two VFXY96 (Verner) forms	12
6. Summary	14
7. Reproducing this comparison	16

1. Overview

EXHALE computes the photoionization rate of each species by integrating its energy-dependent cross section $\sigma(E)$ against the local stellar spectrum on a fixed energy grid (the cross sections are pre-tabulated in `set_energy_vectors.f90`). All cross-section routines live in `src/modules/functions/cross_sec.f90` (module `Cross_sections`) and return σ in units of $1\text{Mb} \equiv 10^{-18}\text{cm}^2$, with photon energy E in eV.

This memo (i) collects the four atomic cross sections relevant to the He 10830 / H α / Ly α diagnostics — H I, He I, He II, and the metastable He I 2³S triplet — and (ii) benchmarks the two *fitted* helium cross sections against independent references:

- **He I ground state:** the two-term ATES fit versus the Verner, Ferland, Korista & Yakovlev (1996; hereafter VFKY96) single-shell analytic fit, over 24.6–500 eV.
- **He I 2³S:** the Norcross-1971 broken power-law fit versus the tabulation of Norcross (1971) data shipped with the p-winds code, over 4.77–60 eV.

Both benchmarks were run against the fits EXHALE carried at the time. They are what settled the two current defaults, and in both cases the fit being benchmarked is the one that was *replaced*: `sigma_HeI` now returns VFKY96 (§4.) and `sigma_HeI23S` the two-VFKY96-wing form (§5.7). The legacy He I fit is still selectable through `ATES_photoionization_rate`; the broken power law is retained only as a comment block.

All numbers and figures below are produced by the companion script `docs/compare_photoion_cross_sections.py` (§7.), which ports the Fortran routines to Python verbatim and writes vector-PDF figures into `docs/figures/`.

2. The four cross sections

2.1 Hydrogenic atoms: H I and He II

H I ($Z = 1$) and He II ($Z = 2$) use the analytic hydrogenic cross section ($\sigma(E, Z)$), with threshold $E_0 = 13.6Z^2$ eV. For $E > E_0$,

$$\sigma(E) = \frac{6.3}{Z^2} \left(\frac{E_0}{E} \right)^4 \frac{\exp[4 - 4 \arctan(\epsilon)/\epsilon]}{1 - \exp(-2\pi/\epsilon)} \text{ Mb}, \quad \epsilon = \sqrt{E/E_0 - 1}, \quad (1)$$

which is the exact non-relativistic result with the Gaunt-factor correction; at threshold $\sigma = 6.3/Z^2$ Mb (6.3 Mb for H I, 1.575 Mb for He II).

2.2 He I ground state (1¹S)

By default the He I ground state (`sigma_HeI`) uses the Verner+1996 (VFKY96) single-shell fit — the form of §2.4 with the He I parameters of Table 1:

$$\sigma_{\text{HeI}}(E) = \sigma_{\text{VFKY96}}(E; E_{\text{th}}=24.59, E_0=13.61, \sigma_0=949.2, y_a=1.469, P=3.188, y_w=2.039, y_0=0.4434, y_1=2.136). \quad (2)$$

Setting `ATES_photoionization_rate: True` in `input.inp` reverts to the legacy ATES two-term fit (valid above $E_{\text{th}} = 24.6$ eV):

$$\sigma_{\text{HeI}}^{\text{ATES}}(E) = \frac{0.6935}{(E/100)^{1.82} + (E/100)^{3.23}} \text{ Mb}. \quad (3)$$

(The 2³S metastable *triplet* is a separate species, Eq. 4 below; Eqs. 2–3 are the 1¹S *singlet* ground state only.)

2.3 He I 2³S metastable (two VFKEY96 forms + bridge)

The metastable triplet — the lower level of He 10830 — (`sigma_HeI23S`) is built from *two* Verner+1996 (VFKEY96, §2.4) single-shell forms joined by a log-linear (power-law) bridge, used piecewise. With $\ell \equiv \log_{10}(E/\text{eV})$ and the node energies $E_3=34.70$, $E_4=45.59$ eV ($x_3 \equiv \log_{10} E_3$, $x_4 \equiv \log_{10} E_4$):

$$\sigma_{\text{He I } 2^3\text{S}}(E) = \begin{cases} \sigma_{\text{VFKEY96}}(E; \text{wing A}) & E \leq E_3 \quad (\text{threshold} \rightarrow \text{Cooper min}) \\ 10^{m_b(\ell-x_3)+y_3^b} & E_3 < E < E_4 \quad (\text{power-law bridge}) \\ \sigma_{\text{VFKEY96}}(E; \text{wing B}) & E \geq E_4 \quad (\text{bump} \rightarrow \text{tail}) \end{cases} \quad (4)$$

Wing A and wing B use the parameters of Table 8 (§5.7; each a $\sim 4\%$ fit to the resonance-averaged cross section), and the bridge is the straight log–log line joining $y_3^b = \log_{10} \sigma_{\text{VFKEY96}}(E_3; A)$ to $y_4^b = \log_{10} \sigma_{\text{VFKEY96}}(E_4; B)$ with $m_b = (y_4^b - y_3^b)/(x_4 - x_3)$. Wing A vanishes below the 4.78 eV threshold (its E_{th}), and wing B supplies the smooth high- E tail ($\sim E^{-7/2}$), so there is no hard cutoff. The construction is continuous at both E_3 and E_4 and reproduces the former broken power law to $\sim 4\%$ (deepest at the bump, 2.57 vs 2.68 Mb).

The *former* five-segment broken power law (Norcross-1971 fit + TOPbase-tail extension of §5.6), retained commented-out in the code, was $\log_{10} \sigma = -0.8134\ell + 1.240$ (4.78–7.49), $-1.772\ell + 2.078$ (7.49–34.70), $9.098(\ell - x_3) - 0.6515$ (34.70–45.59, the bump rise), $-2.660\ell + 4.841$ (45.59–70), and $-2.996\ell + 5.462$ (> 70 eV, the TOPbase tail); $\sigma=0$ below 4.78 eV.

2.4 Verner+1996 single-shell fit (VFKEY96)

This is the VFKEY96 analytic fit, the form EXHALE adopts for metals, for He II¹ and — now by default — for the He I ground state (`sigma_VFKEY96`); for $E \geq E_{\text{th}}$,

$$x = E/E_0 - y_0, \quad z = \sqrt{x^2 + y_1^2}, \quad Q = 5.5 - P/2, \quad (5)$$

$$\sigma(E) = \sigma_0 [(x-1)^2 + y_w^2] z^{-Q} (1 + \sqrt{z/y_a})^{-P} \text{ Mb}. \quad (6)$$

Verner+1996 Table 1 provides ground-shell parameters for *all* ions $Z = 1\text{--}30$, including H I, He I and He II (Table 1); the He I 2³S excited level is *not* in that table, which is why a separate Norcross-based fit is required for the triplet.

Table 1: VFKEY96 ground-shell parameters (Verner+1996 Table 1) used here. σ_0 in Mb.

ion	E_{th}	E_0	σ_0	y_a	P	y_w	(y_0, y_1)
H I	13.60	0.4298	5.475×10^4	32.88	2.963	0.0	(0, 0)
He I	24.59	13.61	949.2	1.469	3.188	2.039	(0.4434, 2.136)
He II	54.42	1.720	1.369×10^4	32.88	2.963	0.0	(0, 0)

3. Overview comparison

Figure 1 shows the four cross sections *as the code evaluates them*: the hydrogenic form for H I and He II, VFKEY96 for the He I ground state, and the two-wing VFKEY96 construction of Eq. (4) for He 2³S. The ordering of thresholds (He 2³S

¹ via the hydrogenic routine

at 4.78, H I at 13.6, He I at 24.6, He II at 54.4 eV) and the peak magnitudes ($\sim 5\text{--}8\text{ Mb}$ near threshold for H I, He I and He 2³S; 1.6 Mb for He II) are evident. The He 2³S curve displays the characteristic Cooper minimum at 34.7 eV, the bridge rising to the resonance-averaged bump at 45.6 eV, and — with no cutoff — the $\sim E^{-3}$ tail of wing B, which carries it below the He II curve by $\sim 100\text{ eV}$.

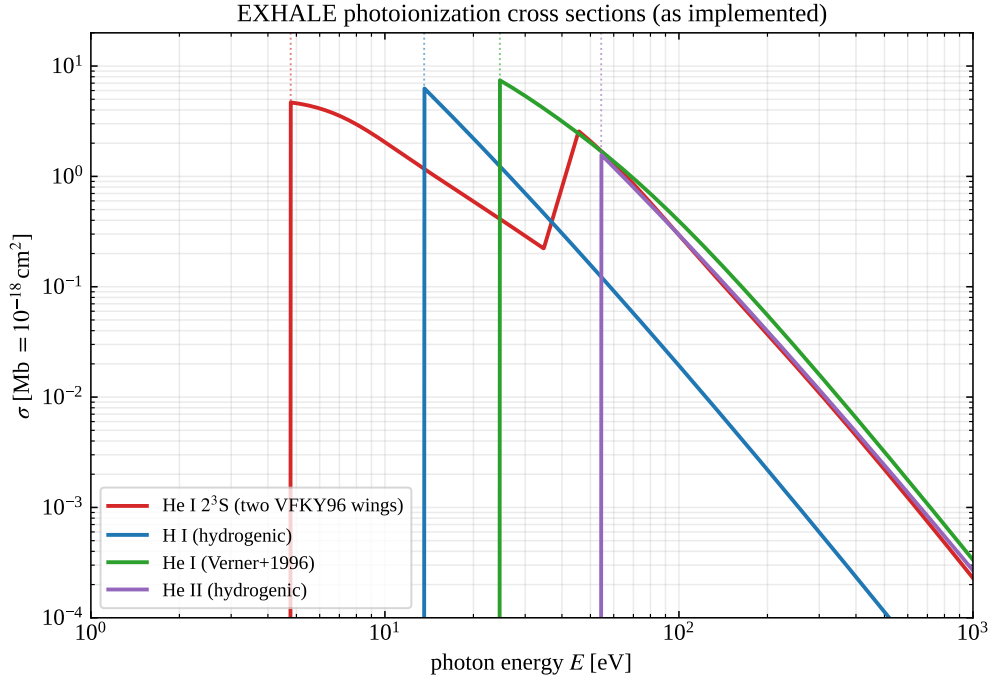


Figure 1: Photoionization cross sections as implemented in EXHALE for H I, He I, He II and the metastable He I 2³S triplet, i.e. with `sigma_HeI` on its VFKY96 default and `sigma_HeI23S` in the two-wing form of Eq. (4). Dotted vertical lines mark the ionization thresholds.

4. He I ground state: the legacy two-term fit vs Verner+1996

This comparison is what settled the default. Because the default `sigma_HeI` is the VFKY96 fit, Figure 2 and Table 2 are best read as what switching `ATES_photoionization_rate: True` on does: the default lies exactly on the VFKY96 reference by construction (the ratio panel is flat at unity), and the two-term fit is the curve that moves. The two agree to within $\sim 14\%$ everywhere over 24.6–500 eV: they match within a few percent at threshold and at high energy, with the two-term form running $\sim 12\text{--}14\%$ below the default in the 50–100 eV range (the band that dominates He photoionization in a typical XUV field). The median ratio over the interval is 0.94.

Because both are accurate fits to the same underlying He I data, the $\leq 14\%$ difference is well within the photoionization-rate uncertainty budget, so this was never a correctness question. It was nevertheless acted on: `sigma_HeI` now calls `sigma_VFKY96` with the He I parameters of Table 1, which unifies He I with the metal and He II treatment and slightly improves the 50–100 eV band. The change moves the He 10830 depth by $\sim 3\%$ (docs/Update_EXHALE §22).

Table 2: He I cross section (Mb): the default sigma_HeI (Verner+1996) and the legacy ATES two-term fit selected by ATES_photoionization_rate: True. The default reproduces the VFKY96 reference to machine precision, so only the option’s ratio is listed.

E [eV]	default (VFKY96)	2-term option	2-term/default
24.6	7.430	7.821	1.053
30	5.361	5.244	0.978
50	2.022	1.779	0.880
100	0.394	0.347	0.880
200	0.055	0.054	0.973
300	0.0160	0.0165	1.028
500	0.0032	0.0035	1.089
ratio over 24.6–500 eV: min 0.861, max 1.089, median 0.939			

5. He I 2³S: the implemented cross section vs p-winds/Norcross

Status of this section. Figure 3 and Tables 3–6 show the *implemented* sigma_HeI23S, i.e. the two-VFKY96-wings-plus-bridge construction of §5.7 and Eq. (4). The subsections §5.1 and 5.4–5.6 are the *derivation record*: their figures deliberately keep plotting the *former* Norcross-1971 five-segment broken power law and the candidate extensions weighed against it, none of which was adopted — the code took the two-wing form of §5.7 instead, and the broken power law survives only as a comment block in cross_sec.f90. Where those subsections say “broken PL”, they mean that former form, not what the code now evaluates.

The p-winds code (dos Santos et al. 2022) ships the He 2³S photoionization cross section as a 23-point table of Norcross (1971), converting a differential oscillator strength $df/d\epsilon$ to a cross section via $\sigma[\text{cm}^2] = 8.0670 \times 10^{-18} (df/d\epsilon)$. Crucially, the node energies E_3, E_4 of Eq. (4), inherited from the breakpoints of the former broken power law, coincide *exactly* with Norcross wavelengths, confirming that sigma_HeI23S descends from a fit to this same data set.

Figure 3 overlays the implemented curve, the tabulated Norcross points, and their log–log interpolation; Table 3 lists the comparison. The agreement is good: the two wings reproduce the table to within $\sim 10\%$ across 4.78–59 eV, with a median ratio of 1.00. The one localized deviation ($\sim 15\%$) sits at the Cooper minimum, 34.7 eV, where the cross section is smallest and therefore least important for the rate. Above 59.2 eV the Norcross table simply ends, so the upper panel of Fig. 3 carries the TOPbase background (§5.1) as the reference instead; wing B follows it to a few percent out to 320 eV (Table 3, last rows).

5.1 Cross-check against TOPbase / Opacity Project

As an independent, fully *ab initio* reference we extracted the He I 2³S (1s2s³S) photoionization cross section from the **TOPbase** (Opacity Project) R-matrix database at the CDS server. The relevant level is NZ = 2, NE = 2, iSLP = 300 (3S, even parity), iLV = 1; the table has 922 points with the photon energy in Rydberg and σ in Mb (level energy -0.35032 Ry , i.e. ionization threshold 4.766 eV). The query URL and parsing are in the companion script (§7.); the raw

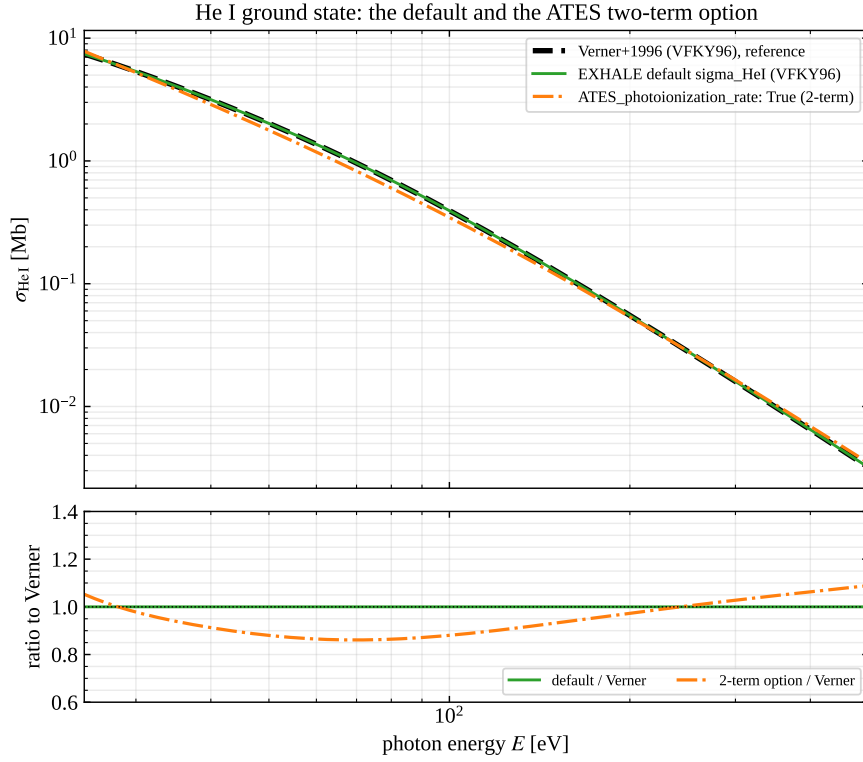


Figure 2: He I ground-state photoionization cross section. The VF KY96 reference (black dashed) is what `sigma_HeI` returns by default (green, lying on it by construction); the legacy ATES two-term fit (orange dash-dot) is what `ATES_photoionization_rate: True` substitutes. Bottom panel: both ratios to VF KY96, so the flat green line is the default and the orange curve is the option's $\pm 14\%$ excursion.

table is saved as `docs/data/topbase_HeI_2_3S_photo.dat`.

Two points must be kept in mind when comparing. (i) The TOPbase threshold sits at the *theoretical* energy (4.63 eV for the first point), ~ 0.14 eV below the experimental 4.77 eV; this is the usual OP threshold offset. (ii) Above ~ 35 eV TOPbase resolves a dense forest of **autoionizing resonances** (doubly-excited states converging on the excited He II core), with peaks reaching $\sigma \simeq 3200$ Mb. These are real but extremely narrow, and the smooth Norcross/EXHALE fits represent only their resonance-averaged background.

Figure 4(a) shows the full range with the resonance structure; panel (b) zooms into the smooth near-threshold region (4.6–25 eV). In that smooth region the three independent descriptions agree to within $\sim 10\%$ (Table 4, which lists the *implemented* wing A): EXHALE/TOPbase has min 0.91 (at the 5 eV threshold edge, where the OP threshold offset dominates), max 1.04, median 1.00 over 5–20 eV. This confirms that wing A — and the Norcross data it descends from — reproduces the OP R-matrix background where it matters for the He 2³S population.

Two caveats applied to the former broken power law: (a) it set $\sigma_{\text{He } 2^3\text{S}} = 0$ above 59.2 eV, whereas TOPbase shows a smooth $\sim E^{-3}$ tail continuing to several hundred eV (small in magnitude, $\lesssim 0.1$ Mb) — wing B of the implemented form supplies that tail, and reproduces TOPbase there to a few percent (0.855 Mb at 70 eV against 0.858, and 0.294 Mb at 100 eV); (b) its “bump” near 45 eV is a smooth stand-in for the resonance complex and should not be read as a

Table 3: He I 2³S cross section (Mb): the implemented two-wing sigma_HeI23S vs p-winds/Norcross. (At threshold 4.78 eV the table interpolation is undefined; the implemented form gives 4.683 Mb against the Norcross threshold value $8.0670 \times 0.605 = 4.881$ Mb. Above 59.2 eV the Norcross table ends; the last rows are the wing-B tail, against which the TOPbase table gives 0.301, 0.0369 and 0.0111 Mb at 100, 200 and 300 eV, and the TOPbase power-law fit of §5.4 gives 0.858 Mb at 70 eV.)

E [eV]	EXHALE	p-winds	ratio
6	4.130	4.092	1.009
8	2.966	3.110	0.953
12	1.466	1.463	1.002
20	0.594	0.546	1.086
30	0.289	0.277	1.042
34.7	0.222	0.262	0.848
45	2.286	2.442	0.936
55	1.650	1.515	1.089
59	1.375	1.251	1.099
70	0.855	—	—
100	0.294	—	—
200	0.0371	—	—
300	0.0110	—	—
ratio over 4.78–59 eV: min 0.849, max 1.100, median 0.999			

true cross section at any single energy there — a caveat the implemented bridge and wing B inherit, since they are resonance-averaged in the same sense.

5.2 Did the > 59 eV truncation matter for the rate?

The relevant quantity for the He 2³S population (and hence He 10830) is the *photoionization rate*, $\Gamma_{2^3S} \propto \int \sigma_{2^3S}(E) N(E) dE$, where $N(E) = J_\nu / h\nu \propto J_{\text{inc}}(E)/E$ is the photon-number flux. In the EXHALE power-law SED, $J_{\text{inc}} \propto E^{\text{PLind}}$ over the EUV band $[e_{\text{low}}, e_{\text{mid}}] = [4.8, 124]$ eV (normalized to L_{EUV}) and the X-ray band $[124, 1240]$ eV (to L_X), so $N(E) \propto E^{\text{PLind}-1}$. Because both $N(E)$ and σ_{2^3S} fall steeply, the integrand peaks at the threshold; the rate is dominated by the near-threshold UV (~ 4.8 – 15 eV), where σ is largest (1–5 Mb) and photons are most numerous.

Table 5 answers this directly with the implemented cross section. Its upper block is the net change of Γ_{2^3S} in going from the former broken power law to Eq. (4), integrated over the whole SED band 4.8–1240 eV; its lower block isolates the share of the implemented rate that comes from $E > 59.2$ eV, the range the broken power law set to zero. The two blocks agree to a few hundredths of a percent, which is the statement that the $\sim 4\%$ reshaping below the bump very nearly cancels in the rate and the tail is the whole of the change. The effect is only 0.1–1.2% across realistic spectral indices (PLind = -1 to -2) and X-ray/EUV ratios; for the EUV slopes typical of cool stars (PLind $\simeq -1.3$ to -1.6) it is

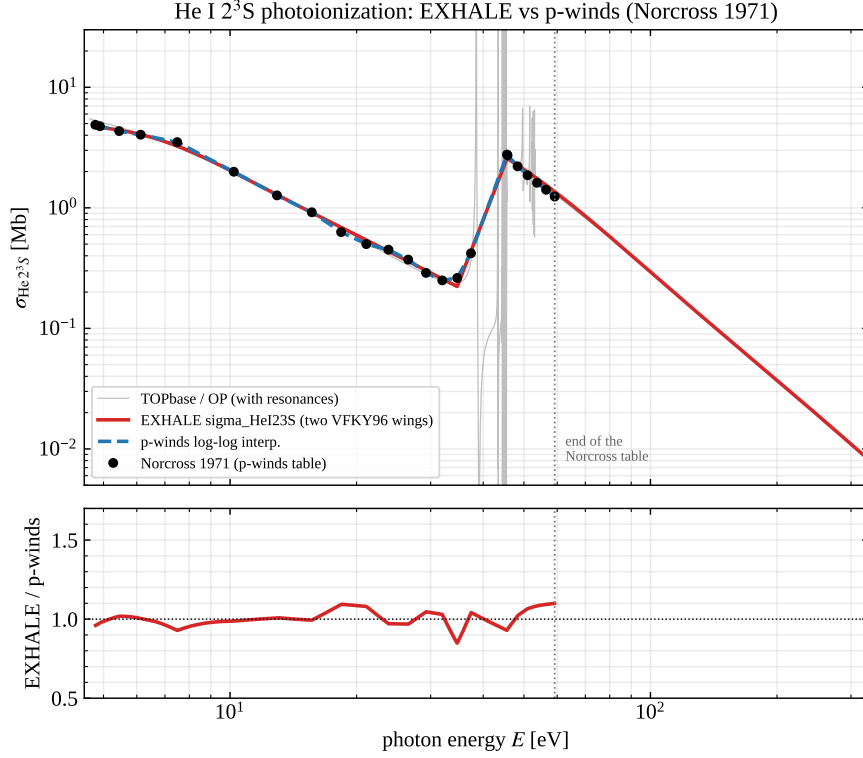


Figure 3: He I 2³S photoionization cross section as implemented (sigma_HeI23S, two VFY96 wings joined by a bridge; red) against the p-winds log–log interpolation (blue dashed) of the Norcross (1971) table (black points), with the TOPbase background (gray) as the reference above 59.2 eV where the Norcross table ends (dotted line). Bottom panel: ratio EXHALE/p-winds over the range the table spans.

0.3–0.6%. This is far below the $\sim 10\%$ uncertainty of the cross-section fit itself (§5.) and of the composition/SED. The 34.7–59.2 eV resonance “bump” likewise carries only $\sim 2\%$ of the rate at $\text{PLind} = -1.3$ (3.3% at -1.0 , 0.5% at -2.0). Moreover, these are *upper bounds*: the power-law SED extrapolates the EUV slope down to 4.8 eV, whereas a real star has a much larger photospheric near-UV flux (4.8–13.6 eV); with a realistic SED the near-threshold region dominates even more and the > 59 eV contribution shrinks further. (Heating is also unaffected: the 2³S number density is negligible compared with the H/He ground states, so its photoionization deposits negligible energy regardless of the high-energy tail.) Removing the cutoff was therefore a cosmetic, sub-percent refinement, which is why it could be folded into the reshaping of §5.7 rather than being worth a change of its own.

5.3 Impact of the autoionizing resonances

Between ~ 40 and 55 eV, TOPbase shows a dense forest of autoionizing resonances (doubly-excited states converging on the He II $n = 2$ threshold at 65.4 eV), with peaks reaching $\sigma \simeq 3223$ Mb, whereas sigma_HeI23S represents this region by a smooth, resonance-averaged bump peaking at only ~ 2.6 Mb (Fig. 4a shows the former broken power law, whose bump peaked at ~ 2.4 Mb; the numbers below use the implemented form). It is tempting to ask whether the resonances, being so tall, substantially raise the He 2³S photoionization rate.

Table 4: He I 2³S cross section (Mb) in the smooth near-threshold region: the implemented sigma_HeI23S (wing A there) vs Norcross/p-winds vs TOPbase.

E [eV]	EXHALE	p-winds	TOPbase	EX/TB	PW/TB
5.0	4.605	4.671	5.081	0.906	0.919
6.0	4.130	4.092	4.231	0.976	0.967
8.0	2.966	3.110	2.866	1.035	1.085
10.0	2.043	2.069	2.025	1.009	1.022
12.0	1.466	1.463	1.488	0.985	0.983
16.0	0.877	0.872	0.877	1.000	0.994
20.0	0.594	0.546	0.574	1.034	0.952
EXHALE/TOPbase over 5–20 eV: min 0.906, max 1.035, median 1.002					

They do not — but the analysis requires care, because *the raw TOPbase table must not be integrated directly*. Integrating both cross sections on the *same* native TOPbase energy grid (so the quadrature is identical) gives the total-rate ratio TOPbase/EXHALE in Table 6. With no clipping the resonances appear to raise the rate by $\sim 14\%$ (at PLind = -1.3); but this increase is produced almost entirely by the 2–4 tallest grid points. The narrowest resonances ($\Delta E \sim 0.01$ eV) are sampled by only one or two points, so the trapezoidal rule assigns them a triangular area fixed by the arbitrary mesh spacing rather than by the physical Lorentzian width. Clipping σ at $\gtrsim 100$ Mb removes just these few unresolved spikes and the ratio collapses to 0.99–1.00: **the robust net effect of the resonances on the rate is below 1%**.

Physically this is expected: although individual resonances are very tall, they are extremely narrow and therefore intercept little photon flux. The resonance-averaged cross section $\langle \sigma \rangle = \sigma_{\text{bg}} + \sum_i (\text{resonance area})_i / \Delta E$ sits only modestly above the smooth background, which the few-Mb Norcross/EXHALE values already approximate (the bridge and wing B of Eq. 4 are exactly such a resonance-averaged representation). The net resonance effect ($< 1\%$) is comparable to the > 59 eV tail (§5.2) and well below the $\sim 10\%$ fit and composition/SED uncertainties, so it does not change the He 10830 conclusions.

Practical caveat: for percent-level work one must use a properly resonance-averaged Opacity-Project cross section (e.g. the resonance-averaged tabulations in the spirit of Bautista et al.); the raw 922-point TOPbase table should never be fed into a broadband rate integral, as the quadrature artifact in Table 6 illustrates.

5.4 A smooth high-energy extension ($E > 59$ eV)

Derivation record. This subsection and the two that follow record the candidates weighed for removing the 59 eV cutoff. Their figures keep plotting those candidates and the former broken power law; the adopted form is §5.7.

Although the > 59 eV truncation was dynamically harmless (§5.2), a smooth analytic extension is trivial to provide and removes the (cosmetic) hard cutoff. Above the He II $n = 2$ resonance limit ($E \gtrsim 66$ eV) the TOPbase cross section

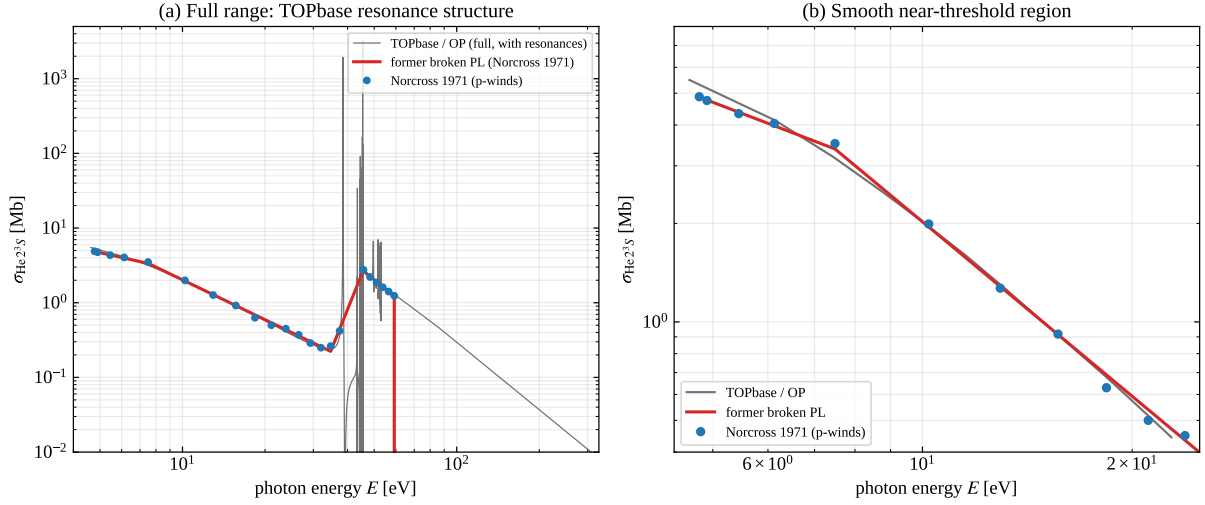


Figure 4: **Derivation record.** He I 2³S photoionization: the *former* broken power law (red), Norcross/p-winds table (blue points), and TOPbase/OP (gray). This is the cross-check that motivated reshaping $\sigma_{\text{HeI}23\text{S}}$; the red curve is the starting point, not what the code now evaluates (Fig. 3). **(a)** Full range: TOPbase resolves autoionizing resonances above ~ 35 eV (peaks to ~ 3200 Mb) that the smooth fits average over; the broken power law is truncated to zero above 59 eV. **(b)** Smooth near-threshold region: the three agree to within $\sim 8\%$ there.

is smooth and monotonic, and a single power law fits the resonance-free background to $\sim 5\%$ over 60–320 eV (Fig. 5):

$$\sigma_{\text{He } 2^3\text{S}}(E) \simeq 0.294 \left(\frac{E}{100 \text{ eV}} \right)^{-2.99} \text{ Mb} = 10^{5.45} E^{-2.99} \text{ Mb} \quad (E > 59 \text{ eV}), \quad (7)$$

with E in eV. The index -2.99 is close to the asymptotic hydrogenic s -shell value ($-7/2$ is approached only at much higher E), and the fit residuals are ≤ 0.023 dex. This is the resonance-free background by construction: it is fit only to points above the $n = 2$ resonance limit and deliberately ignores the 40–55 eV resonance forest (§5.3).

This is also the number the implemented wing B reproduces: it returns 0.294 Mb at 100 eV, so the boxed normalization is what the code now delivers there. Two variants of splicing it onto the former broken power law were weighed at the time — replace the “zero above x_5 ” branch with one more log-linear segment of slope -2.99 : (i) *data-matched* — use the boxed normalization, which reproduces TOPbase to 5% but sits $\sim 16\%$ above the broken-PL endpoint at the 59.2 eV junction (a step at $\sigma \sim 1.4$ Mb, irrelevant to the rate); (ii) *continuity-matched* — anchor at the broken-PL value $\sigma(59.2) = 1.21$ Mb, i.e. $\sigma = 1.21 (E/59.2)^{-2.99}$ Mb, which is continuous but runs $\sim 15\%$ below TOPbase. Either would have been adequate; neither was adopted, because §5.7 replaced the whole construction.

5.5 A unified full-range representation from TOPbase

The power-law tail of §5.4 only patches the high-energy end. A cleaner option is a single representation that follows the TOPbase *resonance-averaged* background over the whole range, 4.77–320 eV, with no ad-hoc junction. Because the cross section has a deep Cooper minimum near 30 eV, a resonance-averaged enhancement (the He II $n = 2$ channel) peaking near 50 eV, and an E^{-3} tail, no single compact analytic form captures it. Instead we tabulate a short (E, σ) node

Table 5: He 2³S photoionization rate for an EXHALE power-law SED of index PLind and X-ray/EUV luminosity ratio L_X/L_{EUV} . *Upper block*: relative change (%) of the rate from the former broken power law to the implemented Eq. (4), integrated over 4.8–1240 eV. *Lower block*: share (%) of the implemented rate carried by $E > 59.2$ eV, the range the broken power law truncated to zero.

PLind	$L_X/L_{\text{EUV}} = 0.1$	0.3	1.0
<i>implemented vs former broken power law</i>			
–1.0	1.16%	1.17%	1.24%
–1.3	0.55%	0.57%	0.63%
–1.6	0.24%	0.25%	0.31%
–2.0	0.04%	0.05%	0.11%
<i>share of the implemented rate from $E > 59.2$ eV</i>			
–1.0	1.16%	1.17%	1.24%
–1.3	0.59%	0.61%	0.66%
–1.6	0.30%	0.31%	0.37%
–2.0	0.12%	0.14%	0.19%

table and interpolate it **log–log with a monotone cubic (PCHIP)** — exactly the scheme EXHALE already uses for the metal cooling fits, so it is a drop-in. Below 30 eV and above 66 eV the nodes are the clean TOPbase background; in 30–66 eV they are the *resonance-averaged* (spike-capped) values, i.e. the sharp resonances are deliberately averaged over rather than followed.

This representation (Fig. 6) reproduces the TOPbase smooth background to within $\sim 10\%$ across the full range (the residual at the very threshold is the ~ 0.14 eV theoretical-threshold offset, §5.1), is C^1 -continuous by construction, and avoids the hard cutoff. Crucially, the He 2³S photoionization *rate* it produces differs from the broken power law by only +1.7% (PLind = –1.3; consistent with §5.2 and §5.3), confirming that the upgrade is a fidelity/ tidiness improvement rather than a change that moves the He 10830 prediction. The 30–66 eV nodes carry the largest uncertainty ($\sim 30\%$, from the resonance averaging) but the least weight.

Not adopted. This node table was the most faithful of the options considered — smooth everywhere, within $\sim 10\%$ of the OP R-matrix background, +1.7% in rate — but it would have introduced an interpolation table where every other cross section in `cross_sec.f90` is a closed form. §5.7 reaches the same shape analytically and is what the code carries; the node table is kept here as the reference the implemented form was checked against.

5.6 A minimal-change extension of the broken power law

The tidiest upgrade of the broken power law keeps essentially all of it and changes only its *last* segment. The procedure is:

1. Keep the broken power law unchanged up to the resonance-averaged bump peak at x_4 (45.6 eV; $\sigma = 2.68$ Mb) — i.e. segments a_1 , a_2 and the rising segment are untouched.

Table 6: Convergence/spike test of the resonance contribution. Total He 2³S rate ratio TOPbase/EXHALE (PLind = −1.3; EXHALE is the implemented Eq. 4), both integrated on the same native grid, as the TOPbase cross section is clipped at an upper value; “pts>cap” counts the grid points removed.

σ clip [Mb]	TOPbase/EXHALE	pts > cap
none (raw)	1.137	0
500	1.026	2
200	1.003	2
100	0.996	4
50	0.992	7

Table 7: Node table for the unified He 2³S photoionization cross section (log–log PCHIP interpolation; $\sigma = 0$ outside the range).

E [eV]	4.77	6	8	10	13.6	20	30	40	52	63	80	100	150	200	300
σ [Mb]	4.88	4.09	3.11	2.07	1.16	0.55	0.273	0.50	1.9	1.14	0.577	0.294	0.085	0.037	0.0111

2. Fit the smooth TOPbase background above $E_J = 70$ eV with a power law:

$$\sigma_{\text{tail}}(E) = 10^{5.462} E^{-2.996} \text{ Mb} \quad (E > 70 \text{ eV}), \quad (8)$$

($\sigma_{\text{TOPbase}}(70 \text{ eV}) = 0.858 \text{ Mb}$; fit residuals $\leq 3\%$).

3. Re-aim the last power-law segment so it runs from the bump peak at x_4 to exactly $\sigma_{\text{tail}}(70 \text{ eV})$ at 70 eV: its slope changes only slightly, from −3.039 to **−2.658** (intercept $c_3 = 4.838$ in $\log_{10} \sigma = a'_3 \log_{10} E + c_3$). Above 70 eV use σ_{tail} ; drop the old 59 eV cutoff.

By construction the result is continuous at *both* joins — at 45.6 eV (with the kept rising segment) and at 70 eV (with the TOPbase tail) — so there is no step and no kink beyond the tiny 0.38-in-slope change. It reproduces the TOPbase background to $\sim 3\%$ above 70 eV and tracks Norcross/p-winds throughout (Fig. 7), while changing the He 2³S photoionization rate by only +0.65% (PLind = −1.3) — the smallest of all the options. This was the smallest of the options — a one-line slope change plus a new tail branch — but it too was superseded: the implemented form of §5.7 lands on the same TOPbase tail (0.855 Mb at 70 eV against the 0.858 this construction was aimed at) while replacing the segmented power law with two Verner forms.

5.7 He 2³S as two VFKEY96 (Verner) forms

It is natural to ask whether the He 2³S cross section could use the same VFKEY96 fits (§2.4) EXHALE already applies to He I and the metals, instead of the bespoke broken power law. Fitting a *single* VFKEY96 term to each smooth part of the curve works very well (Fig. 8, Table 8): segments 1–2 (threshold → Cooper minimum, 4.85–34.7 eV) and segments 4–5 (resonance-averaged bump → tail, 45.7–320 eV) are *each* reproduced to $\sim 4\%$. This is physically sensible — the two terms are the 2s-shell channel and the (resonance-averaged) He II $n = 2$ channel.

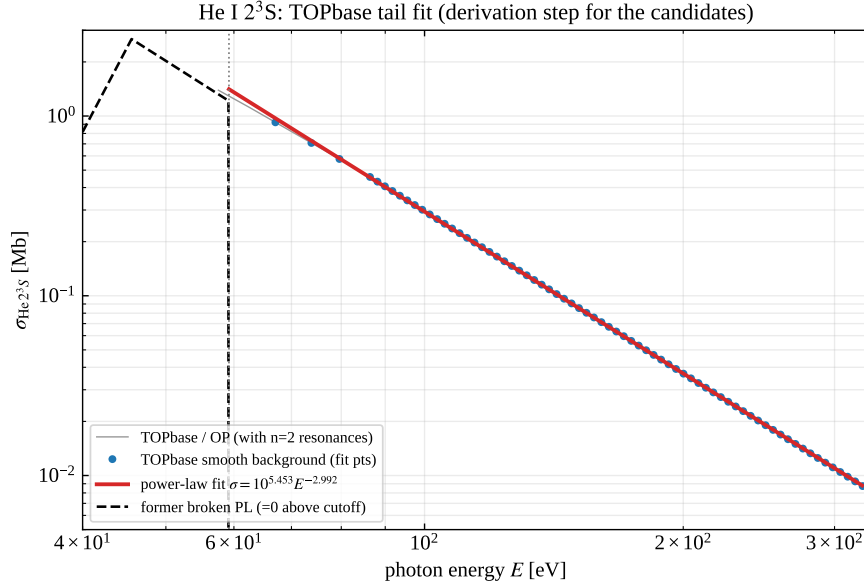


Figure 5: **Derivation record.** TOPbase tail fit for $\sigma_{\text{He } 2^3\text{S}}$, the input to the candidates of §§5.5–5.6. The power-law fit (red) to the resonance-free TOPbase background (blue points, $E \geq 66\text{ eV}$) follows $\sigma \propto E^{-2.99}$ to $\sim 5\%$ up to 320 eV . The former EXHALE broken power law (black dashed) rises to a resonance-averaged bump near 45 eV and is cut to zero above 59.2 eV (dotted line); the implemented wing B follows the red curve instead.

The one region the two wings do not cover is the *transition* ($34.7\text{--}45.6\text{ eV}$, shaded in Fig. 8), where the cross section climbs from the Cooper minimum to the bump. This is bridged by a short log-linear (power-law) segment joining wing A at $E_3 = 34.70\text{ eV}$ to wing B at $E_4 = 45.59\text{ eV}$. **This two-VFKY96-plus-bridge form is now the EXHALE implementation** of $\sigma_{\text{He I } 2^3\text{S}}$ (Eq. 4); it reproduces the former broken power law to $\sim 4\%$ (deepest at the bump, 2.57 vs 2.68 Mb), is continuous at E_3 and E_4 , and reuses the same σ_{VFKY96} routine as He I and the metals. The former broken power law is retained, commented out, in the code.

Table 8: Single-VFKY96 fits to the two smooth parts of He 2^3S (form of §2.4; σ_0 in Mb, energies in eV). Each reproduces its region to $\sim 4\%$. These are the parameters coded in `cross_sec.f90`; re-running the fit reproduces every one of them to better than 0.03% , except y_a of wing A, which is degenerate — at $y_a \sim 10^{12}$ the factor $(1 + \sqrt{z/y_a})^{-P}$ is unity to 10^{-5} over the fitted range, so the refit’s 9.26×10^{11} and the coded 10^{12} give the same cross section.

region	E_{th}	E_0	σ_0	y_a	P	y_w	y_0	y_1
seg 1–2 ($4.85\text{--}34.7$)	4.78	2.645	20.8	1.0×10^{12}	3.42	2.681	1.956	2.603
seg 4–5 ($45.7\text{--}320$)	45.59	49.68	1052	0.0439	2.941	1.717	5.5×10^{-5}	1.118

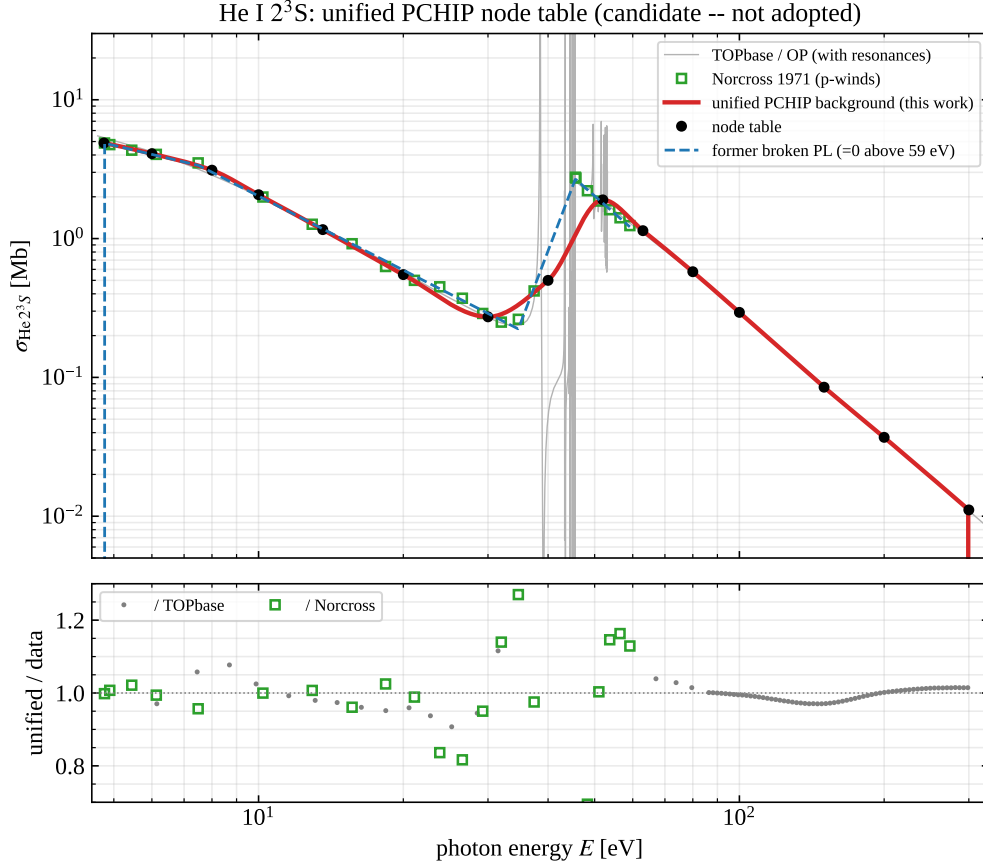


Figure 6: **Candidate, not adopted.** Unified full-range representation (red) of $\sigma_{\text{He I } 2^3\text{S}}$: log–log PCHIP through the node table (black points; Table 7) following the TOPbase resonance-averaged background (gray, with resonances). The Norcross (1971) / p-winds table (green squares) and the former EXHALE broken power law (blue dashed; cut to zero above 59 eV) are overplotted. Bottom: ratio of the unified fit to the TOPbase smooth-background points (gray) and to the Norcross points (green); both within $\sim \pm 15\%$, with the larger Norcross scatter at 40–55 eV reflecting its coarse sampling of the resonance region.

6. Summary

- EXHALE uses three distinct prescriptions: an exact hydrogenic form for H I and He II, Verner+1996 (VFKY96) for the He I ground state (default; legacy two-term fit behind `ATES_photoionization_rate`), and — for the He I 2³S triplet — **two VFKY96 forms joined by a power-law bridge** (§5.7; formerly a Norcross-1971 broken power law, retained commented-out in the code).
- The legacy two-term He I fit agrees with Verner+1996 to within 14% (24.6–500 eV; median 0.94), running slightly low at 50–100 eV. That comparison is why Verner+1996 is now the default.
- The implemented two-wing He I 2³S cross section reproduces the Norcross/p-winds table its shape descends from to within $\sim 10\%$ (4.78–59 eV; median 1.00), including the Cooper minimum at 34.7 eV — where the one $\sim 15\%$ deviation sits — and the bump near 45 eV. It reproduces the former broken power law itself to $\sim 4\%$.

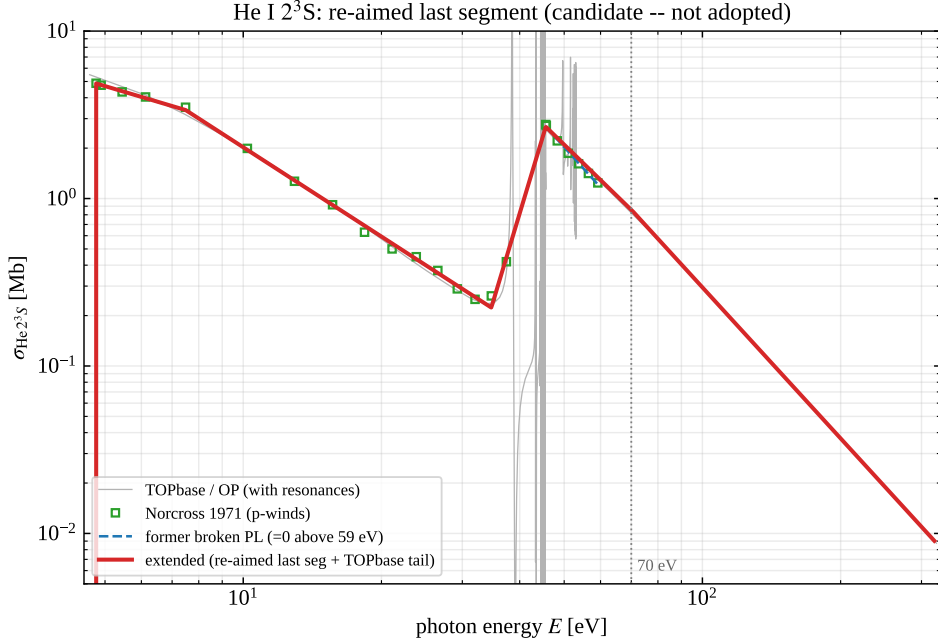


Figure 7: **Candidate, not adopted.** Minimal-change extension (red): the former broken power law up to the 45.6 eV bump, a re-aimed last segment (slope $-3.039 \rightarrow -2.658$) reaching the TOPbase fit at 70 eV (dotted line), and the TOPbase power-law tail above. Norcross/p-winds (green) and TOPbase (gray) are overplotted; the former broken power law (blue dashed) cuts to zero at 59 eV. The curve is continuous at both joins.

- An independent check against the **TOPbase / Opacity Project** R-matrix cross section (§5.1) confirms agreement to $\sim 10\%$ in the smooth near-threshold region (5–20 eV; median 1.00). TOPbase additionally shows autoionizing resonances above ~ 35 eV (averaged over by the fits) and a smooth tail above 59 eV, which the former broken power law truncated to zero and the implemented wing B follows to a few percent.
- The 40–55 eV autoionizing resonances change the He 2^3S rate by less than 1% once the few numerically under-resolved spikes are excluded (§5.3); the raw TOPbase table must not be integrated directly.
- Three ways of removing the 59 eV cutoff were measured and none was adopted on its own: a $E^{-2.99}$ power-law tail (§5.4), a 15-node log–log PCHIP table (+1.7% in rate, §5.5), and a re-aimed last segment meeting $10^{5.462} E^{-2.996}$ Mb at 70 eV (+0.65%, §5.6). The implemented two-wing form (§5.7) delivers the same tail analytically: it returns 0.855 Mb at 70 eV and 0.294 Mb at 100 eV.
- Both fitted helium cross sections are adequate for the photoionization balance. Extending the He 2^3S fit above 59 eV changes its photoionization rate by only 0.1–1.2% (§5.2), so the former cutoff was harmless; He I has since been migrated to the VFKY96 form (Table 1) for internal consistency, which moves the He 10830 depth by $\sim 3\%$.

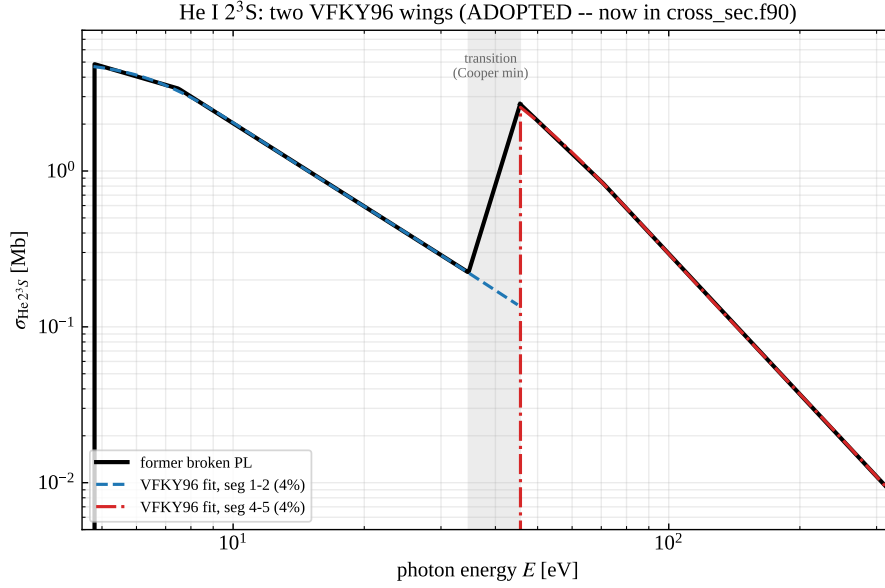


Figure 8: **Adopted.** He 2³S (black: the former broken power law + TOPbase tail) with single-VFKY96 fits to segments 1–2 (blue dashed) and 4–5 (red dash-dot), each good to ~4%. These two fits are wing A and wing B of Eq. (4) and are what cross_sec.f90 now carries. The shaded band (34.7–45.6 eV) is the Cooper-minimum/rise transition where the two must be spliced piecewise; a simple sum fails there.

7. Reproducing this comparison

All figures and numbers are generated by docs/compare_photoion_cross_sections.py:

```
cd EXHALE_v1.00/docs
python3 compare_photoion_cross_sections.py
# -> figures/xsec_overview.pdf
#   figures/xsec_HeI_vs_Verner.pdf
#   figures/xsec_HeI23S_vs_pwinds.pdf
#   figures/xsec_HeI23S_topbase.pdf
#   figures/xsec_HeI23S_tail.pdf
#   figures/xsec_HeI23S_unified.pdf
#   figures/xsec_HeI23S_ext.pdf
#   figures/xsec_HeI23S_vfky.pdf
# reads: data/topbase_HeI_2_3S_photo.dat
```

The TOPbase table was retrieved (once) from the CDS Opacity Project server; topbase.sh below is that server’s CGI endpoint, not a script of this repository. The query was

```
curl -L -G "https://cds.unistra.fr/cgi-bin/topbase/topbase.sh" \
  --data "com=dt" --data "ent=p" \
```

```
--data "nz1=2" --data "nz2=2" --data "ne1=2" --data "ne2=2" \  
--data "is1=3" --data "is2=3" --data "il1=0" --data "il2=0" \  
--data "ip1=0" --data "ip2=0" --data "lv1=1" --data "lv2=1" \  
--data "en1=0.0" --data "en2=0.0" --data "so=s1"
```

and parsed into data/topbase_HeI_2_3S_photo.dat (photon energy in Ry $\rightarrow$ eV; σ in Mb).

The script ports sigma and sigma_VFKY96 from cross_sec.f90 (using the same constants h_{ev} , c from parameters.f90), hard-codes the Verner+1996 He I/HI/He II parameters and the 23-point p-winds/Norcross table, and emits the figures listed above. Its sigma_HeI and sigma_HeI23S are the *former* Fortran routines — the two-term ATES fit and the five-segment broken power law — which is what this memo benchmarks; it has not been re-pointed at the current cross_sec.f90, so re-running it reproduces the tables and figures as published rather than the shipped cross sections.